

## Cultivars

Plant breeders are constantly trying to improve a plant's performance, and produce new "cultivars" (a contraction of **cultivated varieties**) that are more vigorous, produce more flowers, or have other favorable characteristics that differentiate them from the wild form of the species.

Although many cultivars are bred by specialists, others are found in the wild or occur as mutations and are then introduced to cultivation. To reproduce cultivars that exhibit a consistent set of characteristics, many must be propagated vegetatively (cuttings, grafting, or division) or grown annually from specially selected seed. Cultivars have vernacular names, which are printed in Roman type within quotes (eg. *Phygelius aequalis* 'Yellow Trumpet').

When plant breeders raise a new cultivar, it is given a code name that may be different from the name under which the plant is sold. For example, the rose Casino also has the code name 'Macca'; in this book, both names are cited, and styled: *Rosa* CASINO ('Macca').

## Hybrids

Sexual crosses between botanically distinct species or genera are known as hybrids and are indicated by a multiplication sign. If the cross is between species in different genera, the result is called an "intergeneric hybrid" and, when two or more genera are crossed, the name given is a condensed form of the the relevant genera; x *Cuprocyparis*, for example, covers hybrids between species of *Chamaecyparis* and *Cupressus*.

If more than three genera are involved, then the hybrids are named after a person and given the ending "-ara". Thus x *Potinara*, a hybrid of *Brassavola*, *Cattleya*, *Laelia*, and *Sophronitis*, commemorates M. Potin of the French orchid society. More common are "interspecific hybrids," which are crosses between species in the same genus. These have a collective name preceded by a multiplication sign: *Epimedium* x *rubrum* covers hybrids between *E. alpinum* and *E. grandiflorum*.

When a plant is grafted onto another, a new plant occasionally arises that contains tissues of both parents. These are named in the same way as sexual hybrids, but are denoted by a plus sign: + *Laburnocytisus* 'Adamii', for example, is a graft hybrid between *Laburnum* and *Chamaecytisus*.

Cultivars of hybrids are listed under a botanical name, or if the parentage is complex, by giving the generic name followed solely by the cultivar name (eg. *Rosa* 'Buff Beauty').

## Visual Key to Plant Classification

In horticulture, plants are classified according to a hierarchical system and named primarily on the basis of Linnaeus's binomial approach (genus followed by species epithet). As an example, part of the Rosaceae family is set out below, showing all levels of this system.

### FAMILY

A group of several genera that share a set of underlying natural characteristics. Family names usually end in *-aceae*. Family limits are often controversial.

### GENUS (PL. GENERA)

A group of one or more plants that share a range of distinctive characteristics. Several (rarely one) genera are classified into one family. Each genus contains one or more species and its name is printed in italic type with an initial capital letter.

### SPECIES

A group of plants that breeds naturally to produce offspring with similar characteristics; these keep it distinct from other populations in nature. Each species has a two-part name printed in italic type.

### SUBSPECIES

A naturally occurring, distinct variant of a species, differing in one or more characteristic. Indicated by "subsp." in Roman type and an epithet in italic type.

### VARIETAS AND FORMA

A *varietas* (var.) is a minor species subdivision, differing slightly in botanical structure. A *forma* (f.) is a minor variant of a species, often differing in flower color or habit from others in the species.

### CULTIVAR

Selected or artificially raised, distinct variant of a species, subspecies, *varietas*, *forma*, or hybrid. Indicated by a vernacular name printed in Roman type within single quotation marks.

